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A Project Phase I(BCD685) Report on

“Deepfake Detection in Low-Resolution KYC Videos Using rPPG and Hybrid Quantum Machine Learning”

A Dissertation work submitted in partial fulfillment of the requirement for the award of the degree

Bachelor of Engineering
in
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2025-2026



Certificate

This is to certify that the **Project Phase I (BCD685)** entitled “**Deepfake Detection in Low-Resolution KYC Videos Using rPPG and Hybrid Quantum Machine Learning**” carried out by **Ishaan Choudhary (1AY23CD025)**, **Jhashank Nayan(1AY23CD027)**, **P Aswin Krishna (1AY23CD041)** and **Sumit Kumar Gupta (1AY23CD060)**, are bonafide students of **Acharya Institute of Technology, Bengaluru** in partial fulfillment for the award of the degree of **Bachelor of Engineering in Computer Science and Engineering (Data Science)** of the **Visvesvaraya Technological University, Belagavi** during the year **2025-2026**. It is certified that all corrections/suggestions indicated for Internal Assessment have been incorporated in the report deposited in the departmental library. The Project Phase I report has been approved as it satisfies the academic requirements in respect of Project Phase I prescribed for the said degree.

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ABSTRACT

Deepfake technology has rapidly evolved with the advancement of generative artificial intelligence, creating a major threat to digital trust in financial Video Know Your Customer (KYC) systems. Financial institutions are increasingly exposed to manipulated identity verification videos, especially in low-resolution environments where compression noise, weak lighting, motion blur, and unstable facial regions reduce the effectiveness of traditional detection methods.

This project focuses on developing a deepfake video detection framework for low-resolution KYC environments using remote photoplethysmography (rPPG) and hybrid quantum machine learning. The proposed system follows a two-stage architecture in which rPPG-based physiological analysis acts as the primary evidence layer, while hybrid quantum-classical classification functions as the final decision-making stage.

The system processes uploaded KYC videos through frame extraction, frame quality assessment, face detection, tracking, alignment, low-resolution preprocessing, and region-of-interest extraction. Temporal physiological signals are then extracted from facial skin regions such as cheeks and forehead to evaluate pulse consistency and biological rhythm behavior. The generated features are converted into compact feature vectors and supplied to a hybrid quantum machine learning classifier for final discrimination between genuine and manipulated videos.

The project emphasizes low-resolution robustness rather than benchmark-quality media assumptions. Instead of depending entirely on visible image artifacts, the proposed framework studies whether a face behaves like a real living face over time through physiological consistency analysis. The final system produces confidence-aware outputs such as real, fake, or uncertain, making the architecture suitable for practical financial verification workflows.

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CHAPTER 1

INTRODUCTION

1.1 Overview

Deepfakes are AI-generated synthetic media that manipulate facial appearance using deep learning techniques. Their rapid growth has created major concerns in identity verification systems, especially in financial video KYC environments where user authentication depends on facial verification through low-resolution videos.

The proposed project focuses on building a deepfake video detection system for financial KYC applications where the input videos may contain compression noise, weak lighting, motion blur, unstable face regions, and bandwidth limitations. The system is designed specifically for low-resolution environments where traditional forensic cues become unreliable.

The proposed framework introduces a two-stage verification architecture. The first stage uses remote photoplethysmography (rPPG) for physiological signal analysis, while the second stage uses hybrid quantum machine learning for final classification. The architecture combines preprocessing, facial tracking, ROI extraction, temporal signal analysis, feature conditioning, and confidence-aware classification into a unified framework.

1.2 Motivation

Financial institutions increasingly depend on remote video verification systems for customer onboarding and identity validation. However, manipulated deepfake videos can bypass conventional authentication mechanisms and support identity fraud. Existing systems mainly focus on visual artifact analysis and perform poorly under low-resolution conditions.

The project is motivated by the need to:

- Improve fraud prevention in financial verification systems.
- Develop robust low-resolution deepfake detection mechanisms.

- Use physiological consistency instead of only texture analysis.
- Explore hybrid quantum machine learning for subtle feature classification.
- Support confidence-aware decision-making in practical KYC workflows.

1.3 Objective

The objectives of the proposed project are:

- To extract video frames and perform standardization for reliable model input.
- To analyze frame sequences and extract facial cues using tracking and consistency checks.
- To incorporate rPPG-based liveness detection for physiological authentication.
- To integrate a hybrid quantum-classical decision layer for improved classification accuracy and authentication output.
- To support low-resolution financial KYC verification environments.

1.4 Scope

The project focuses specifically on deepfake detection in low-resolution KYC video systems used in financial verification workflows. The architecture is not designed as a general-purpose social media deepfake detector but as a secure identity verification framework.

The scope of the project includes:

- Low-resolution video preprocessing.
- Face detection and alignment.
- rPPG-based physiological analysis.
- Temporal feature extraction.
- Hybrid quantum-classical classification.
- Confidence-aware verification outputs.

Limitations include:

- Dependence on facial visibility.
- Sensitivity of physiological extraction to motion and lighting conditions.

- Experimental nature of hybrid quantum learning frameworks.

1.5 Existing System

Existing deepfake detection systems include CNN-based models, CNN-GRU hybrid models, transformer-based systems, and multimodal audio-visual architectures. These methods mainly depend on visual artifacts, temporal inconsistencies, or multimodal fusion for classification.

Major existing approaches include:

1. CNN-Based Detection Systems

- Use frame-level texture analysis.
- Limited temporal awareness.
- Weak performance under low-resolution conditions.

2. Hybrid CNN-GRU Architectures

- Capture spatial and temporal information.
- Require stable sequential inputs.
- Limited generalization capability.

3. rPPG-Based Physiological Systems

- Analyze biological pulse-related signals.
- Sensitive to lighting and motion variations.

4. Multimodal Transformer Systems

- Combine audio and visual signals.
- High computational complexity.
- Limited real-time applicability.

1.6 Proposed System

The proposed system introduces a **physiologically grounded deepfake detection framework** designed specifically for **low-resolution financial KYC video environments** using **remote photoplethysmography (rPPG)** and **hybrid quantum machine learning** techniques. The architecture follows a structured **two-stage verification pipeline** that combines physiological signal analysis with advanced hybrid classification for reliable deepfake detection under compressed and noisy video conditions.

Stage 1: rPPG-Based Physiological Analysis

In the first stage, the system extracts **temporal pulse-related physiological signals** from facial skin regions such as the cheeks and forehead using rPPG-based analysis. The extracted signals are analyzed to evaluate **biological consistency, temporal rhythm stability, pulse synchronization behavior, and liveness-related characteristics** across consecutive video frames. This stage focuses on identifying whether the observed facial behavior corresponds to a genuine human physiological pattern or an artificially manipulated deepfake sequence.

Stage 2: Hybrid Quantum Machine Learning

In the second stage, the generated physiological and temporal features are converted into **compact normalized feature representations** and supplied to a **hybrid quantum-classical machine learning classifier**. The classification layer performs final discrimination between genuine and manipulated videos by learning subtle physiological inconsistencies and low-resolution temporal patterns that are difficult to capture using conventional visual-forensic approaches alone.

Key Features of the Proposed System

- Robust deepfake detection under **low-resolution and compressed KYC video conditions**.
- Physiological liveness verification using **rPPG-based temporal biological signal analysis**.

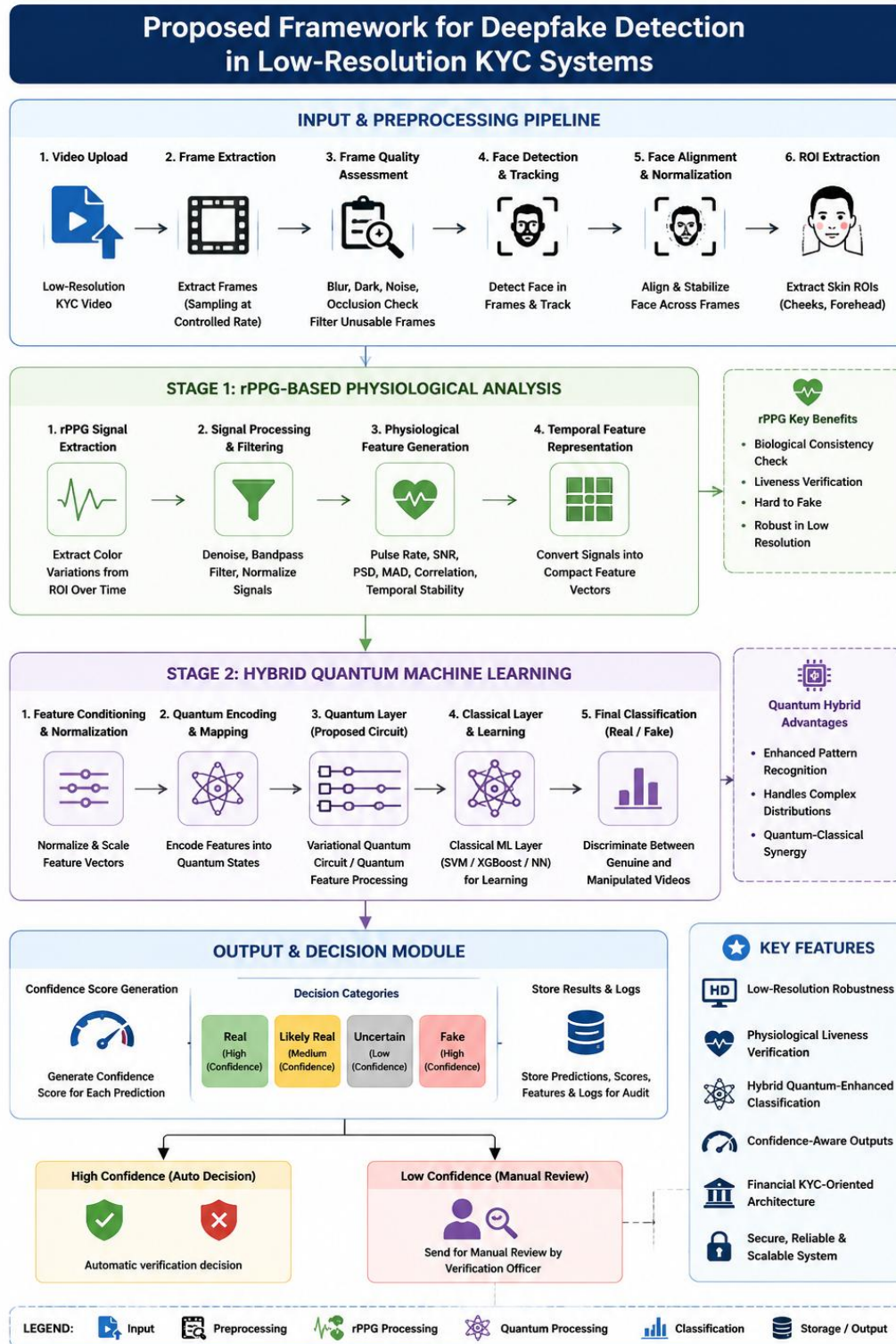


Figure 1.1 - Proposed Framework for Deepfake Detection in Low-Resolution KYC Systems

CHAPTER 2

PROBLEM STATEMENT

2.1 Problem Statement

Detecting deepfake videos in low-resolution financial KYC environments is a highly challenging task due to factors such as poor video quality, compression artifacts, unstable facial regions, motion blur, weak lighting conditions, and limited availability of reliable forensic features. Traditional deepfake detection approaches primarily depend on visible texture inconsistencies and spatial visual artifacts, but these cues become significantly weaker and less reliable when videos are heavily compressed or captured under degraded real-world conditions.

In practical financial verification systems, uploaded KYC videos are often recorded using mobile devices operating under bandwidth limitations and unstable environmental conditions. As a result, the facial information available for authentication may contain noise, inconsistent illumination, facial jitter, and temporal distortion that negatively affect conventional image-forensics-based detection models. Existing methods that focus only on frame-level visual analysis frequently fail to maintain reliable performance when applied to low-resolution KYC videos.

Financial institutions require robust identity authentication mechanisms capable of operating reliably under real-world low-quality video environments without increasing false acceptance or false rejection rates. A false acceptance may support identity fraud and unauthorized access, while a false rejection may negatively affect legitimate customer verification workflows. Therefore, there is a strong requirement for a physiologically grounded deepfake detection system capable of analyzing biological consistency rather than depending entirely on visible appearance artifacts.

The proposed work addresses this challenge by combining remote photoplethysmography (rPPG)-based physiological verification with hybrid quantum-classical machine learning techniques to improve deepfake detection robustness in low-resolution financial KYC systems.

2.2 Motivation

The primary motivation of the project is to enable secure and reliable financial KYC verification by preventing identity fraud and reducing false authentication decisions in poor-quality video environments. Existing visual-artifact-based approaches become unreliable under degraded low-resolution conditions, making it necessary to explore alternative evidence sources that remain stable even when visual quality is weak.

The project focuses on physiological realism instead of depending entirely on facial texture artifacts. Biological signals such as pulse-related temporal color variations are difficult to reproduce consistently in manipulated videos, making rPPG-based analysis a promising approach for practical liveness verification. The proposed framework therefore emphasizes temporal physiological consistency as a stronger indicator of authenticity in comparison to traditional appearance-based methods.

The motivation for the project is further strengthened by the rapid increase in deepfake-enabled financial fraud attempts and the growing use of AI-generated synthetic identities in online verification systems. Real-world KYC environments additionally introduce challenges such as low bandwidth, compression noise, unstable lighting, and inconsistent facial visibility, which further weaken conventional detection pipelines.

The project is further motivated by the following factors:

- Increasing number of deepfake-based fraud attempts targeting financial identity verification systems.
- Weakness and instability of traditional image-forensics approaches under low-resolution and compressed video conditions.
- Need for confidence-aware authentication mechanisms capable of supporting practical financial verification workflows.
- Growing importance of physiological liveness verification using rPPG-based temporal signal analysis.
- Emerging research opportunities in hybrid quantum-classical machine learning for advanced deepfake classification.
- Requirement for robust and scalable verification systems suitable for real-world financial applications.

2.3 Objectives

The main objective of the proposed work is to design and develop a robust deepfake detection framework for low-resolution financial KYC environments using physiological signal analysis and hybrid quantum machine learning techniques. The proposed system aims to improve authentication reliability by combining temporal biological verification with advanced classification mechanisms.

The objectives of the proposed work are as follows:

- To implement low-resolution video preprocessing techniques capable of preserving important temporal and physiological information under degraded video conditions.
- To perform frame extraction, facial localization, alignment, and region-of-interest selection from low-resolution KYC videos.
- To extract remote photoplethysmography (rPPG)-based physiological signals from facial skin regions such as cheeks and forehead.
- To analyze temporal pulse consistency, biological synchronization behavior, and physiological rhythm stability for deepfake identification.
- To generate compact physiological feature representations suitable for hybrid machine learning-based classification.
- To develop a hybrid quantum-classical classification framework for final discrimination between genuine and manipulated videos.
- To generate confidence-aware authentication outputs such as real, fake, and uncertain predictions for practical verification support.
- To support secure and scalable financial KYC verification workflows operating under real-world low-resolution conditions.

Prepared using only the information and terminology present in your uploaded project documents and review material.

CHAPTER 3

DETAILED LITERATURE SURVEY

1. A Study on Deepfake Detection Methods

Methodology

This paper presents a comprehensive survey-based methodology for analyzing deepfake detection approaches. The authors reviewed traditional machine learning methods, deep learning techniques such as CNNs and GAN-based forensic systems, and blockchain-assisted detection mechanisms. The study compares multiple detection approaches based on their working principles, robustness, accuracy, and applicability in domains like healthcare, entertainment, and politics. The methodology mainly focuses on literature synthesis rather than implementing a new model. The researchers categorized existing deepfake detection systems and analyzed their effectiveness against modern synthetic media manipulation techniques.

Research Gap

The study identifies that existing deepfake detection systems still suffer from poor generalization across datasets and unseen manipulation methods. Many approaches work only in controlled environments and fail under low-resolution, compressed, or adversarial conditions. The paper also highlights the lack of standardized benchmark datasets and limited real-time deployment capabilities. Another major research gap is the absence of unified multimodal frameworks combining spatial, temporal, physiological, and frequency-based forensic analysis.

2. Real-Time Deepfake Video Detection using Machine Learning: A CNN-based Authentication Framework

Methodology

This paper proposes a CNN-based real-time deepfake detection framework optimized for computational efficiency and high accuracy. The model was trained using approximately 15,000 images and incorporated feature extraction, adaptive loss functions, and real-time processing optimizations. The framework detects texture inconsistencies, manipulation artifacts, and abnormal facial patterns in videos. The proposed system was evaluated against state-of-the-art models such as EfficientNet and Xception and demonstrated robustness against adversarial attacks and video compression artifacts.

Research Gap

Although the framework achieves strong performance, it still struggles to generalize across unseen deepfake generation methods. Traditional CNN architectures are limited in capturing subtle temporal inconsistencies and complex manipulations introduced by modern GANs and diffusion models. The framework also lacks multimodal learning capabilities and transformer-based contextual understanding. Scalability for large-scale deployment and improved robustness against evolving deepfake attacks remain major research challenges.

3. Multimodal Vision Transformer Forensics for Deepfake Resistant Secure Authentication**Methodology**

This paper proposes a multimodal transformer-based forensic framework called MViT-Forensics. The system combines spatial RGB embeddings extracted through Vision Transformers or Swin Transformers, frequency-domain residuals obtained using DCT/FFT analysis, and physiological rPPG signals extracted through temporal attention encoders. These multimodal features are fused using a cross-attention transformer architecture. The framework also integrates an anomaly-aware intrusion scoring mechanism to generate authenticity confidence scores. Experiments were conducted using FaceForensics++, Celeb-DF, DFDC Preview, and MMDF datasets.

Research Gap

Despite achieving high detection accuracy, the framework requires heavy computational resources because of multimodal transformer processing. Real-time deployment on low-resource devices remains difficult. The model also suffers from limited explainability due to the complexity of transformer-based fusion mechanisms. Another challenge is maintaining robustness against future unseen deepfake generation techniques, severe video compression, lighting variations, and occlusions.

4. Utilizing rPPG Signal Synchronization and Deep Learning Techniques for Deepfake Video Detection**Methodology**

This paper introduces Sync_rPPG, a physiological signal-based deepfake detection framework. The methodology extracts remote photoplethysmography (rPPG) signals from facial regions, especially the left and right cheeks, and evaluates synchronization using Signal-to-Noise Ratio (SNR), Power Spectral Density (PSD), Pearson Correlation Coefficient (PCC), Mean Absolute Deviation (MAD), and Standard Deviation (SD). Discrete Wavelet Transform (DWT) is used for time-frequency decomposition. The framework combines CNN and LSTM architectures to capture both spatial and

temporal inconsistencies in manipulated videos.

Research Gap

The framework faces challenges in extracting stable rPPG signals from low-quality, compressed, or low-light videos. Physiological-signal-based detection methods are highly sensitive to motion blur, facial occlusion, makeup, and head movement. Another limitation is the dependency on high-quality face tracking and temporal consistency. The system also lacks broader generalization to non-facial or highly distorted deepfake content.

5. Quantum-Enhanced Deepfake Detection: A Hybrid Classical-Quantum Deep Learning Approach

Methodology

This paper proposes a hybrid quantum-classical deepfake detection architecture integrating Xception CNN, Vision Transformer (ViT), and Quantum Convolutional Layers implemented using PennyLane. CNN and transformer components extract local and global features, while quantum convolutional operations enhance feature representation capabilities. The framework also uses focal loss, mixup augmentation, and test-time augmentation to improve robustness and address class imbalance.

Research Gap

Although the model achieves high performance, practical implementation remains difficult because scalable quantum hardware is not yet widely available. The architecture also introduces significant computational complexity and high training costs. Another research gap is the absence of benchmark datasets specifically designed for evaluating hybrid quantum-classical deepfake detection systems.

6. Deepfake Detection using Multi-Stream Convolutional Neural Networks with Attention-based Fusion

Methodology

This paper proposes a Multi-Stream CNN framework for robust deepfake detection. The methodology includes face detection, alignment, frequency-domain analysis, residual noise mapping, rPPG signal extraction, and optical flow analysis. Multiple CNN streams independently process spatial, spectral, and temporal features. An attention-based adaptive fusion mechanism dynamically combines the extracted features before classification using a softmax layer.

Research Gap

The proposed architecture is computationally expensive due to multimodal processing and multiple parallel CNN streams. Real-time deployment on edge devices and mobile platforms remains difficult.

The framework also heavily depends on accurate preprocessing techniques such as face alignment and optical flow estimation. Generalization to unseen deepfake techniques is still a major challenge.

7. Robust Deepfake Detection for Electronic Know Your Customer Systems Using Registered Images

Methodology

This paper introduces a deepfake detection framework specifically designed for e-KYC systems. The methodology evaluates temporal inconsistencies in identity vectors extracted using face recognition models. It compares embeddings from input videos with embeddings from registered authentic images to identify identity mismatches. The framework is capable of detecting both face swapping and face reenactment attacks while maintaining robustness against image degradation.

Research Gap

The framework depends heavily on the availability of authentic registered images, which may not always be feasible in real-world applications. Another limitation is reduced robustness against highly sophisticated spoofing attacks and severe image degradation conditions. The paper also identifies the need for privacy-preserving authentication mechanisms and lightweight deployment strategies.

8. Two-Branch Deepfake Detection Network Based on Improved Xception

Methodology

This paper proposes a two-branch deepfake detection architecture consisting of a global branch and a local branch. The global branch uses an improved Xception network combined with GRU for complete video analysis, while the local branch processes frame-level artifacts using augmentation techniques such as Face-Cutout. The model is evaluated on FaceForensics++ and Celeb-DF datasets for cross-dataset and low-resolution detection performance.

Research Gap

Although the framework improves low-resolution detection performance, it still struggles with highly compressed and noisy videos. The two-branch architecture increases computational complexity and training time. Another challenge is improving robustness against advanced GAN-generated manipulations and cross-domain deepfake attacks.

9. EffiSwinNet: A Cross-Architecture Deepfake Detection with StyleGAN Generated Faces

Methodology

This paper proposes EffiSwinNet, a hybrid deepfake detection framework combining EfficientNetB0 and Swin Transformer architectures. The system uses shifting window transformer blocks for efficient

contextual learning while EfficientNet extracts local spatial features. The model was trained using a dataset containing approximately 140K real and fake StyleGAN-generated images. The hybrid architecture achieved higher performance compared to standalone Swin Transformer models.

Research Gap

The framework mainly focuses on image-level deepfake detection and does not effectively address temporal inconsistencies in videos. Transformer-based architectures also require high computational resources, limiting real-time deployment. Another research gap is the lack of multimodal forensic analysis combining physiological, temporal, and frequency-domain features.

10. An Overview of Deepfake Video Detection Using Remote Photoplethysmography (rPPG)

Methodology

This review paper studies deepfake detection approaches based on remote photoplethysmography (rPPG). The methodology analyzes how physiological signals generated from subtle skin color variations can be used to distinguish real and manipulated videos. The paper reviews GAN-based and Autoencoder-based deepfake generation techniques and evaluates spatial, frequency-domain, and physiological-based detection approaches. It also discusses signal extraction methods and the role of deep learning algorithms in identifying inconsistencies in blood flow patterns.

Research Gap

The paper identifies that existing rPPG-based approaches struggle under poor illumination, motion blur, heavy video compression, and low-resolution conditions. Another major limitation is the lack of standardized datasets specifically designed for physiological-signal-based deepfake detection. The study also highlights the absence of hybrid systems that combine rPPG analysis with spatial and temporal forensic features.

11. A Quantum-Hybrid Framework for Enhanced Deepfake Detection: Integrating QAOA-Based Feature Selection With Quantum-Inspired Attention Mechanisms

Methodology

This paper proposes a hybrid quantum-classical deepfake detection framework combining InceptionResNetV1, CNNs, QAOA-based feature selection, and Quantum-Inspired Attention Mechanisms. The methodology uses InceptionResNetV1 to extract high-dimensional features from images. Quantum Approximate Optimization Algorithm (QAOA) is then applied for selecting the most informative features. A multi-head quantum-inspired attention layer dynamically reweights important features before classification. Focal loss with label smoothing is used to address class imbalance, and

Grad-CAM visualization is integrated for explainability.

Research Gap

Although the framework improves feature selection and classification performance, it still suffers from high computational complexity and scalability issues. Practical implementation using real quantum hardware remains difficult due to technological limitations. Another limitation is the lack of lightweight quantum-inspired architectures suitable for real-time applications and mobile deployment.

12. Short and Low Resolution Deepfake Video Detection Using CNN

Methodology

This paper proposes a CNN-based framework specifically designed for detecting deepfake videos in short-duration and low-resolution conditions. The methodology uses the DFDC and FaceForensics++ datasets and applies facial feature extraction using convolutional neural networks to identify manipulation artifacts. The framework is optimized to handle compressed and low-quality videos that are commonly found on social media platforms.

Research Gap

The model still struggles to detect highly sophisticated manipulations under extreme compression and poor lighting conditions. CNN-based architectures also have limited capability in learning long-term temporal dependencies. Another research gap is the lack of multimodal analysis combining spatial, temporal, and physiological forensic features.

13. Hybrid Recurrent Deep Learning Model for DeepFake Video Detection

Methodology

This paper introduces multilayer hybrid recurrent deep learning models for deepfake video detection. The methodology combines temporal facial convolutional features with recurrent architectures such as LSTM and GRU to capture motion inconsistencies and temporal anomalies in videos. The proposed system focuses on learning sequential video features and comparing hybrid recurrent networks with traditional stacked recurrent architectures.

Research Gap

The framework mainly focuses on temporal learning while giving limited importance to frequency-domain and physiological forensic analysis. Training recurrent architectures on long video sequences also increases computational complexity and training time. The model further lacks robustness against adversarial attacks and unseen deepfake generation techniques.

14. Design and Analytical Performance of a Hybrid Model Using Deep Learning Techniques for

the Detection of Deepfake Videos

Methodology

This paper investigates hybrid CNN-GRU architectures for deepfake video detection. The methodology integrates GRU layers with EfficientNet-B1, InceptionV3, DenseNet121, and Xception models. Both standalone CNN architectures and CNN-GRU hybrid architectures are evaluated to study the role of temporal information in identifying manipulated videos. The GRU layers are responsible for capturing sequential dependencies and temporal inconsistencies between video frames.

Research Gap

Although temporal learning improves detection accuracy, the framework remains computationally expensive for real-time deployment. Another challenge is reduced generalization across unseen datasets and newly emerging GAN-based manipulations. The paper also lacks explainability techniques and multimodal fusion strategies.

15. EffiSwinNet: A Cross-Architecture Deepfake Detection with StyleGAN Generated Faces

Methodology

This paper proposes EffiSwinNet, a hybrid architecture combining EfficientNetB0 and Swin Transformer for detecting StyleGAN-generated deepfake images. EfficientNet extracts local spatial features while the Swin Transformer captures global contextual information using shifting window transformer blocks. The model was trained on a dataset containing approximately 140K real and fake images and compared with standalone transformer-based models.

Research Gap

The framework mainly focuses on image-based deepfake detection and does not effectively handle temporal inconsistencies in videos. Transformer-based architectures also demand high computational resources, making deployment on lightweight systems difficult. Another limitation is the absence of multimodal forensic analysis involving physiological and frequency-domain information.

16. Two-Branch Deepfake Detection Network Based on Improved Xception

Methodology

This paper proposes a two-branch architecture for deepfake detection. The global branch uses an improved Xception network combined with GRU to analyze full videos, while the local branch processes frame-level artifacts using data augmentation methods such as Face-Cutout. The framework aims to improve cross-dataset and low-resolution deepfake detection performance using FaceForensics++ and Celeb-DF datasets.

Research Gap

Although the model improves low-resolution detection accuracy, it still faces challenges with highly compressed and noisy videos. The dual-branch architecture also increases computational complexity and memory requirements. Another research gap is improving robustness against advanced GAN and diffusion-model-generated deepfakes.

17. Deepfake Detection using Multi-Stream Convolutional Neural Networks with Attention-based Fusion**Methodology**

This paper proposes a Multi-Stream CNN architecture that processes multiple modalities simultaneously. The methodology includes face alignment, residual noise mapping, optical flow estimation, frequency-domain analysis, and rPPG signal extraction. Multiple CNN streams independently process spatial, spectral, and temporal information. An attention-based fusion mechanism dynamically combines the extracted features before classification.

Research Gap

The framework is computationally expensive because of multimodal processing and multiple parallel CNN streams. Real-time implementation on edge devices remains difficult. The model also heavily depends on accurate preprocessing techniques such as face alignment and optical flow computation. Generalization against unseen manipulation methods remains limited.

18. Robust Deepfake Detection for Electronic Know Your Customer Systems Using Registered Images**Methodology**

This paper introduces a deepfake detection framework designed specifically for electronic Know Your Customer (e-KYC) systems. The methodology evaluates temporal inconsistencies in identity vectors extracted using face recognition models. It compares facial embeddings from input videos with embeddings from registered authentic images to identify identity mismatches. The system is capable of detecting both face swapping and face reenactment attacks and is designed to remain robust against image degradation.

Research Gap

The framework depends heavily on the availability of authentic registered images, which may not always be available in real-world scenarios. Another challenge is maintaining robustness against highly

advanced spoofing attacks and degraded input conditions. The system also lacks privacy-preserving mechanisms and lightweight deployment strategies for mobile-based authentication systems.

19. Quantum-Enhanced Deepfake Detection: A Hybrid Classical-Quantum Deep Learning Approach

Methodology

This paper proposes a hybrid classical-quantum deepfake detection architecture that combines Xception CNN, Vision Transformer (ViT), and Quantum Convolutional Layers implemented using PennyLane. The methodology integrates CNN-based local feature extraction with transformer-based global contextual understanding to improve deepfake detection accuracy. Quantum convolutional operations are introduced to enhance feature representation and improve classification performance. The framework also applies focal loss, mixup augmentation, and test-time augmentation to improve robustness and address class imbalance. The model was evaluated using deepfake and real-image datasets and achieved high ROC-AUC and F1-score performance.

Research Gap

Although the framework demonstrates strong detection capability, practical implementation remains difficult due to the limited availability of scalable quantum hardware. The hybrid architecture also introduces significant computational complexity and high training costs. Another research gap is the lack of benchmark datasets specifically designed for evaluating hybrid quantum-classical deepfake detection systems. The model also requires further validation against unseen deepfake generation techniques and real-world adversarial attacks.

20. A Study on Deepfake Detection Methods

Methodology

This paper presents a survey-based analysis of existing deepfake detection approaches. The methodology involves reviewing traditional machine learning techniques, deep learning-based CNN methods, GAN-forensic approaches, and blockchain-assisted detection frameworks. The study categorizes detection methods based on spatial analysis, temporal analysis, and forensic feature extraction. The paper also compares the strengths and weaknesses of different detection systems across applications such as healthcare, politics, entertainment, and digital security.

Research Gap

The paper identifies that many existing deepfake detection systems lack robustness against newly emerging GAN and diffusion-based manipulations.

CHAPTER 4

SURVEY SUMMARY TABLE

S. No	Paper Title	Methodology	Key Techniques	Research Gap
1	A Study on Deepfake Detection Methods	Survey and comparative analysis	CNN, GAN, Blockchain	Lack of unified robust systems
2	CNN-based Authentication Framework	Real-time CNN detection	CNN, adaptive loss	Poor generalization
3	MViT-Forensics	Multimodal transformer framework	ViT, Swin, rPPG	High computational complexity
4	Sync_rPPG	Physiological signal synchronization	rPPG, CNN-LSTM DWT	Sensitive to lighting and motion
5	Quantum-Enhanced Detection	Hybrid quantum-classical model	Xception, ViT, Quantum layers	Quantum hardware limitations
6	MS-CNN with Attention Fusion	Multi-stream multimodal CNN	Attention fusion, optical flow	Expensive computation
7	e-KYC Deepfake Detection	Identity inconsistency analysis	Face embeddings, temporal analysis	Dependence on registered images

8	Two-Branch Xception	Global-local branch detection	Xception, GRU	Heavy computational cost
9	EffiSwinNet	Hybrid CNN-transformer model	EfficientNet, Swin Transformer	Limited video-level analysis
10	rPPG Deepfake Overview	Survey of physiological detection	rPPG, GAN, AE	Lack of hybrid frameworks
11	Quantum-Hybrid Framework	QAOA-based feature selection	QAOA, CNN, attention	Scalability issues
12	Low-Resolution CNN Detection	CNN-based low-quality detection	CNN, DFDC, FF++	Weak temporal learning
13	Hybrid Recurrent Model	Temporal recurrent learning	CNN, LSTM, GRU	Limited multimodal analysis
14	Hybrid GRU-CNN Model	CNN-GRU hybrid framework	EfficientNet, GRU	High computational cost
15	EffiSwinNet	CNN-transformer hybrid model	EfficientNet, Swin Transformer	Limited video analysis
16	Two-Branch Xception	Global-local branch detection	Xception, GRU	Heavy computation
17	MS-CNN with Attention Fusion	Multimodal CNN fusion	Optical flow, attention	Expensive computation

18	e-KYC Deepfake Detection	Identity inconsistency detection	Face embeddings	Dependence on registered images
19	Quantum-Enhanced Deepfake Detection	Hybrid quantum-classical framework	Xception, ViT, Quantum Layers	Quantum hardware limitations
20	A Study on Deepfake Detection Methods	Literature survey and analysis	CNN, GAN, Blockchain	Lack of unified robust systems

Table 4.1 - Survey Summary Table

CHAPTER 5

SYSTEM REQUIREMENT SPECIFICATION

5.1 Functional Requirements

- The system should accept low-resolution KYC video inputs captured from financial verification environments and validate whether the uploaded video is readable, properly formatted, and suitable for further processing.
- The system should perform frame extraction from uploaded videos and conduct frame quality assessment to identify blurred, dark, noisy, or unusable frames before analysis.
- The system should detect facial regions from extracted frames and perform face tracking and alignment to maintain stable facial positioning throughout the temporal sequence.
- The system should extract remote photoplethysmography (rPPG)-based physiological signals from facial skin regions such as cheeks and forehead for biological consistency analysis.
- The system should generate temporal physiological features and compact feature representations from the extracted rPPG signals for further classification processing.
- The system should apply a hybrid quantum-classical machine learning framework to classify videos as genuine or manipulated based on the generated physiological and temporal features.
- The system should produce confidence-aware authentication outputs such as real, fake, or uncertain to support secure and practical financial KYC verification workflows.

5.2 Non-functional Requirements

- The system should maintain reliable performance under low-resolution video conditions involving compression noise, weak lighting, and unstable facial regions commonly present in KYC environments.
- The preprocessing pipeline should remain stable and consistent while preserving important physiological and temporal information required for accurate rPPG signal analysis.
- The architecture should support scalability and modular expansion so that additional preprocessing, feature extraction, or classification modules can be integrated in future stages.

- The system should support confidence-aware verification mechanisms to reduce false acceptance and false rejection during financial identity authentication.
- The system should ensure secure handling and processing of sensitive KYC video data throughout all stages of preprocessing, analysis, classification, and result generation.

5.3 Hardware Requirements

- Processor: Intel Core i5, AMD Ryzen 5, or higher processor capable of supporting machine learning preprocessing and video analysis operations efficiently.
- RAM: Minimum 8 GB RAM is required for preprocessing, frame extraction, feature generation, and model execution under experimental conditions.
- Storage: Minimum 50 GB available storage space is required for storing datasets, extracted frames, processed outputs, trained models, and experimental results.
- GPU: NVIDIA GPU support is optional but recommended for accelerating deep learning training, feature extraction, and hybrid model experimentation.
- Internet Connection: A stable internet connection is required for dataset access, framework installation, dependency management, and research-related updates.

5.4 Software Requirements

- **Operating System**
 - Windows 10 or Windows 11 operating system for development, preprocessing, and experimental implementation.
 - Ubuntu Linux environment for machine learning experimentation, framework compatibility, and model execution support.
- **Programming Languages**
 - Python programming language for preprocessing, rPPG extraction, feature engineering, model development, and experimental evaluation.
 - JavaScript for interface-level integration and optional frontend interaction support.
- **Libraries and Frameworks**
 - OpenCV for video processing, frame extraction, image normalization, and face-related preprocessing operations.

- TensorFlow for deep learning model development, training, and classification experimentation.
- PyTorch for neural network implementation and temporal feature learning experiments.
- Qiskit for hybrid quantum machine learning implementation and quantum-enhanced classification experiments.
- NumPy for numerical computation, matrix operations, and feature vector manipulation.
- Scikit-learn for preprocessing utilities, feature conditioning, evaluation metrics, and baseline machine learning models.
- **Development Tools**
 - Visual Studio Code (VS Code) for project development, debugging, and code management.
 - Jupyter Notebook for experimentation, visualization, preprocessing validation, and model analysis.
 - GitHub for version control, project collaboration, and source code management.
- **Dataset**
 - FaceForensics++ Dataset for training, preprocessing, validation, and experimental evaluation of deepfake detection performance under different conditions.

CHAPTER 6

SYSTEM DESIGN

6.1 System Design

The proposed deepfake detection framework is designed as a structured multi-layer architecture optimized for low-resolution financial KYC environments. The system follows a sequential processing pipeline where each layer performs a specific operation before passing refined information to the next stage. The architecture is designed to gradually reduce noise, stabilize facial information, preserve temporal consistency, and improve classification reliability without damaging weak physiological clues required for rPPG analysis.

The overall architecture consists of eight major processing layers, namely the Input Acquisition Layer, Frame Sampling and Quality Assessment Layer, Face Detection and Alignment Layer, Low-Resolution Preprocessing Layer, rPPG Feature Generation Layer, Feature Conditioning Layer, Hybrid Quantum Classification Layer, and Decision Support Layer. Each layer contributes to improving robustness against compression artifacts, unstable lighting conditions, facial motion, and low-resolution degradation commonly observed in practical KYC video systems.

6.1.1 System Architecture

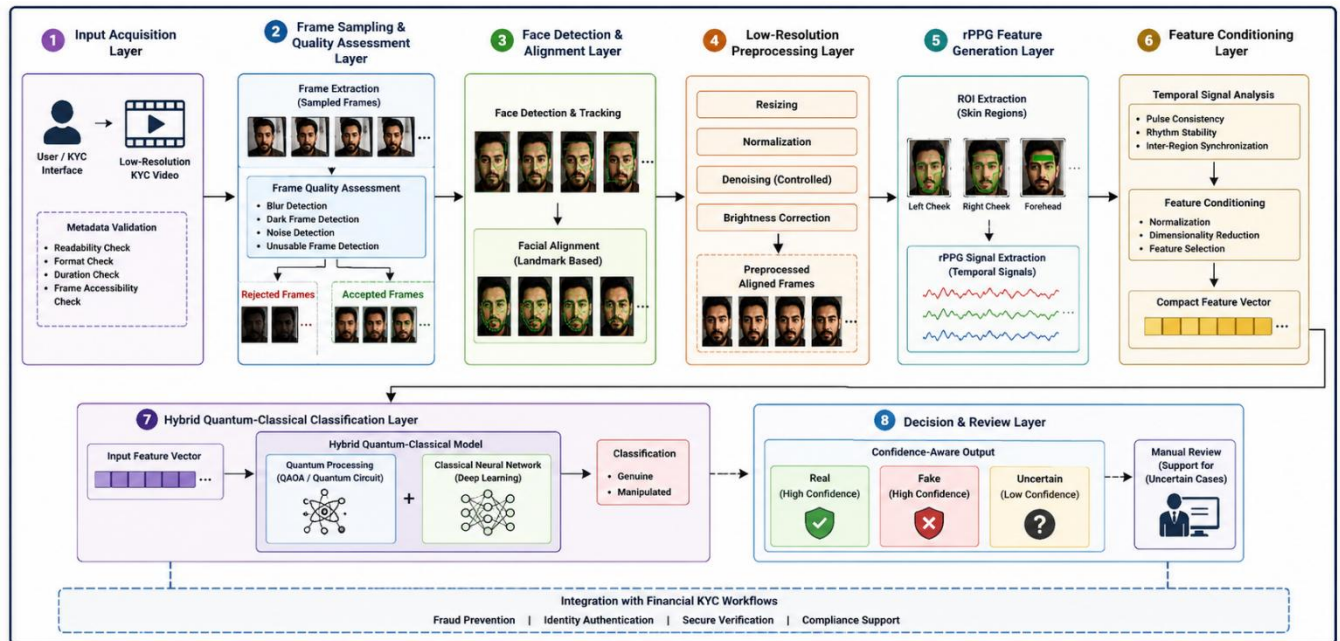
The proposed system architecture begins with the acquisition of low-resolution KYC video input from the user or financial verification interface. During the initial stage, the uploaded video undergoes metadata validation to verify video readability, supported format, duration, and frame accessibility. After successful validation, the video is decomposed into frame sequences at a controlled sampling rate suitable for temporal analysis.

The extracted frames are then passed through a frame quality assessment pipeline where blurred, dark, noisy, or unusable frames are identified and filtered. The remaining frames are processed through face detection and alignment stages to maintain stable facial positioning across consecutive frames. Facial landmarks are utilized to reduce motion-induced inconsistencies and maintain temporal continuity for physiological analysis.

Following facial stabilization, the system performs low-resolution preprocessing operations including resizing, normalization, controlled denoising, and brightness correction while preserving original temporal information. Region-of-interest (ROI) extraction is then performed to isolate skin-rich facial regions such as cheeks and forehead, which are important for remote photoplethysmography (rPPG)-based physiological signal extraction.

The extracted temporal physiological signals are analyzed to generate compact feature representations describing pulse consistency, temporal rhythm stability, and inter-region synchronization behavior. These features are then conditioned and normalized before being supplied to the hybrid quantum-classical classification layer. The classification stage performs final discrimination between genuine and manipulated videos using compact physiological evidence generated from the earlier stages.

Finally, the system produces confidence-aware outputs such as real, fake, or uncertain, which can be integrated into practical financial KYC verification workflows for fraud prevention and identity authentication support.



6.1 – System Architecture

6.1.2 Module Design

Input Acquisition Module

The Input Acquisition Module is responsible for accepting low-resolution KYC videos from users or verification systems. The module validates video readability, file structure, frame accessibility, and metadata properties such as duration, frame rate, and resolution before processing begins. Invalid or corrupted video files are rejected during this stage to prevent unreliable downstream analysis.

Frame Sampling and Quality Assessment Module

This module extracts frames from uploaded videos at a controlled sampling rate suitable for temporal physiological analysis. It additionally performs frame quality evaluation by identifying blurred frames, low-brightness frames, excessive motion distortion, and unusable facial regions. The module ensures that only stable and usable frames are forwarded to later processing stages.

Face Detection and Tracking Module

The Face Detection Module identifies facial regions from extracted video frames and tracks facial movement throughout the temporal sequence. The module focuses on maintaining consistent facial localization while minimizing background interference and unstable cropping behavior.

Face Alignment Module

The Face Alignment Module stabilizes facial landmarks such as eyes, nose, and mouth across consecutive frames. Alignment improves temporal consistency and reduces motion-induced distortion that can negatively affect physiological signal extraction and temporal analysis.

Low-Resolution Preprocessing Module

The Low-Resolution Preprocessing Module standardizes all facial crops into a fixed input format suitable for physiological signal extraction and classification. The preprocessing stage includes resizing, normalization, mild denoising, and brightness stabilization while preserving weak biological and temporal information required for accurate rPPG analysis.

ROI Extraction Module

The Region-of-Interest (ROI) Extraction Module isolates skin-rich facial areas such as the left cheek, right cheek, and forehead for physiological signal analysis. ROI consistency across frames is maintained to improve the stability of temporal physiological feature extraction.

rPPG Signal Extraction Module

This module performs remote photoplethysmography-based analysis by extracting pulse-related temporal color variation patterns from the selected facial regions. The generated physiological signals are analyzed for rhythm stability, temporal synchronization, and biological consistency behavior.

Feature Conditioning Module

The Feature Conditioning Module converts raw physiological outputs into compact and normalized feature vectors suitable for machine learning classification. The module may include normalization, dimensionality reduction, feature selection, and temporal consistency conditioning operations.

Hybrid Quantum Classification Module

The Hybrid Quantum Classification Module acts as the final decision-making stage of the architecture. Conditioned physiological features are processed using a hybrid quantum-classical machine learning framework to classify the input video as genuine or manipulated. The module focuses on learning subtle discriminative patterns from low-resolution physiological evidence.

Decision Support Module

The Decision Support Module converts classification outputs into confidence-aware verification results such as real, fake, likely fake, likely real, or uncertain. This stage supports practical financial verification workflows by enabling manual review support for uncertain authentication cases.

6.2 Detailed Design

The detailed design section describes the structural and behavioral representation of the proposed system using diagrams and interaction models. These diagrams provide a conceptual understanding of

the system workflow, module interactions, entity relationships, and operational sequence involved in low-resolution deepfake detection using rPPG and hybrid quantum machine learning.

6.2.1 Class Diagram

The class diagram represents the structural organization of the proposed system and describes the interaction between major processing components involved in the deepfake detection pipeline. The architecture includes classes such as VideoInput, FrameExtractor, FaceDetector, ROIExtractor, rPPGAnalyzer, FeatureConditioner, QuantumClassifier, and DecisionEngine.

The VideoInput class handles uploaded KYC video management and metadata validation. The FrameExtractor class manages frame decomposition and temporal sequencing operations. The FaceDetector and ROIExtractor classes are responsible for facial localization and physiological region selection respectively. The rPPGAnalyzer class extracts and processes temporal physiological signals from facial skin regions. FeatureConditioner prepares normalized feature vectors for classification, while the QuantumClassifier performs hybrid quantum-classical decision-making.

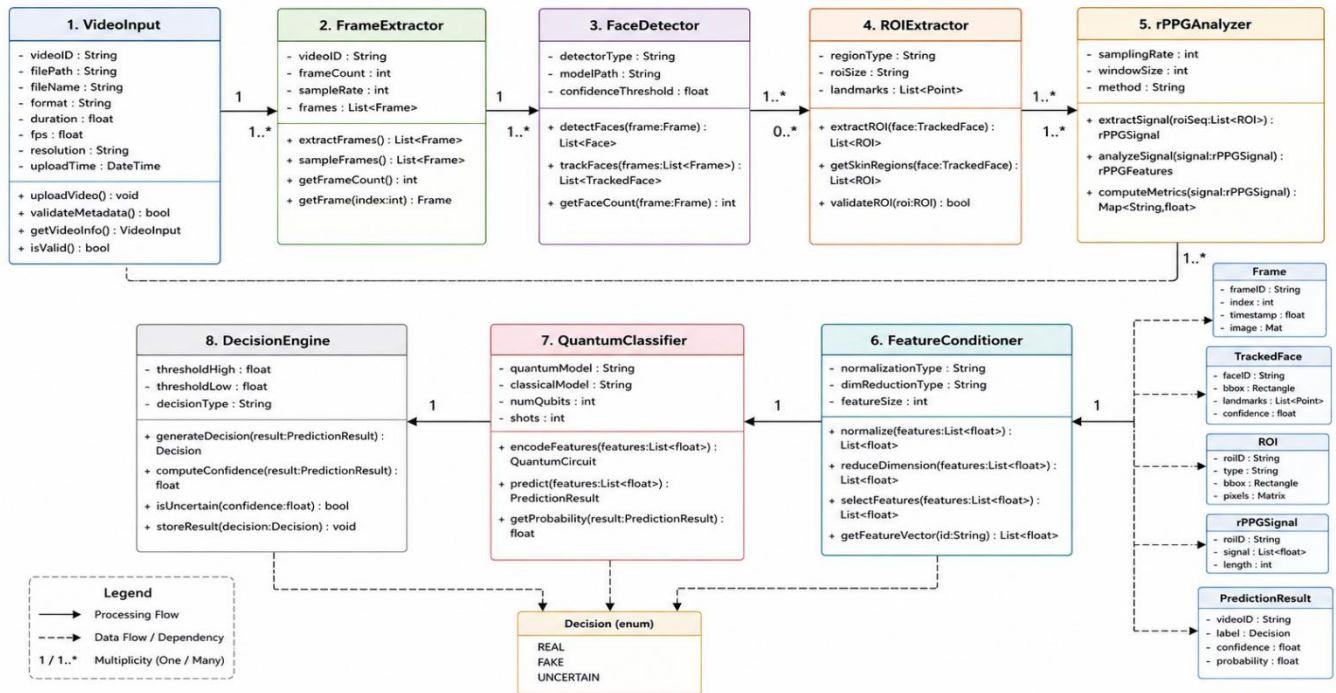


Figure 6.2 – Class Diagram

6.2.2 Activity Diagram

The activity diagram illustrates the operational workflow followed by the proposed deepfake detection framework from video input acquisition to final prediction generation. The process begins when a user uploads a low-resolution KYC video into the system. The uploaded video is validated and decomposed into frames before frame quality assessment is performed.

After removing unusable frames, facial regions are detected and aligned to maintain stable temporal positioning. ROI extraction is then performed to isolate physiological regions for rPPG analysis. The extracted temporal physiological signals are processed into compact feature representations and passed through the hybrid quantum-classical classifier. Finally, the system generates classification outputs and confidence-aware authentication decisions.

The activity flow additionally supports rejection or manual review for uncertain predictions where confidence levels are insufficient for automatic verification.

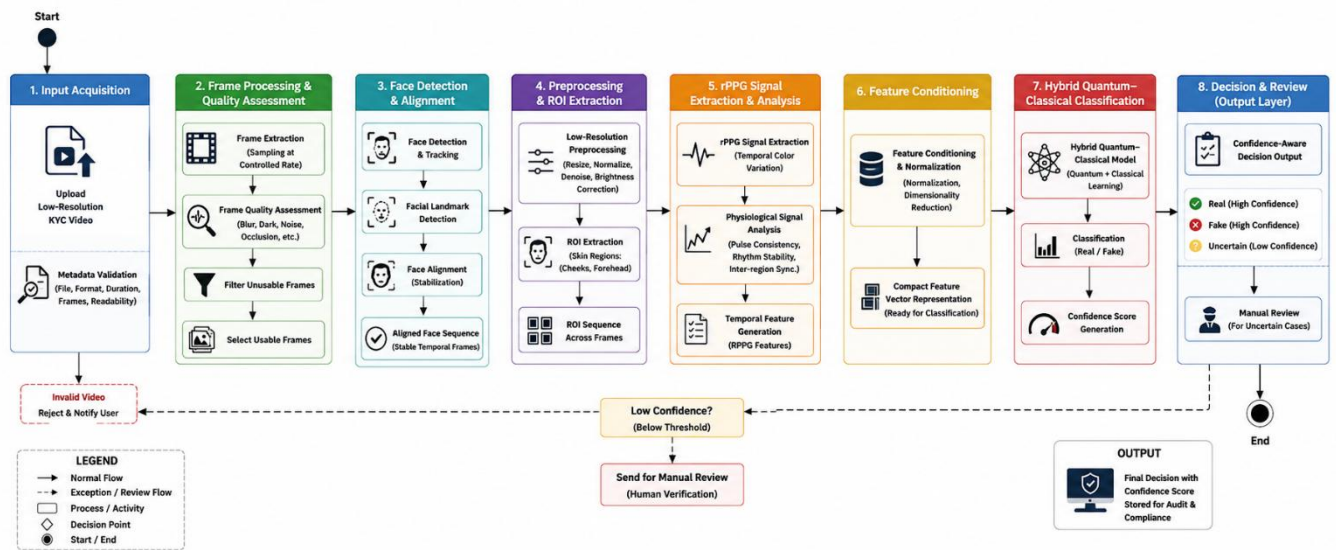


Figure 6.3 – Activity Diagram

6.2.3 Use Case Diagram

The use case diagram describes the interaction between system actors and the major functionalities provided by the proposed deepfake detection framework. The primary actors involved in the system are the User, Financial Verification Officer, and System Administrator.

The User interacts with the system by uploading KYC videos for authentication analysis. The Financial Verification Officer reviews confidence-aware outputs, handles uncertain verification cases, and monitors fraud-related predictions generated by the system. The System Administrator manages system configuration, preprocessing settings, dataset maintenance, and operational monitoring activities.

Major use cases of the system include uploading KYC videos, performing facial preprocessing, extracting physiological signals, generating hybrid classification outputs, producing confidence scores, and reviewing uncertain authentication cases.

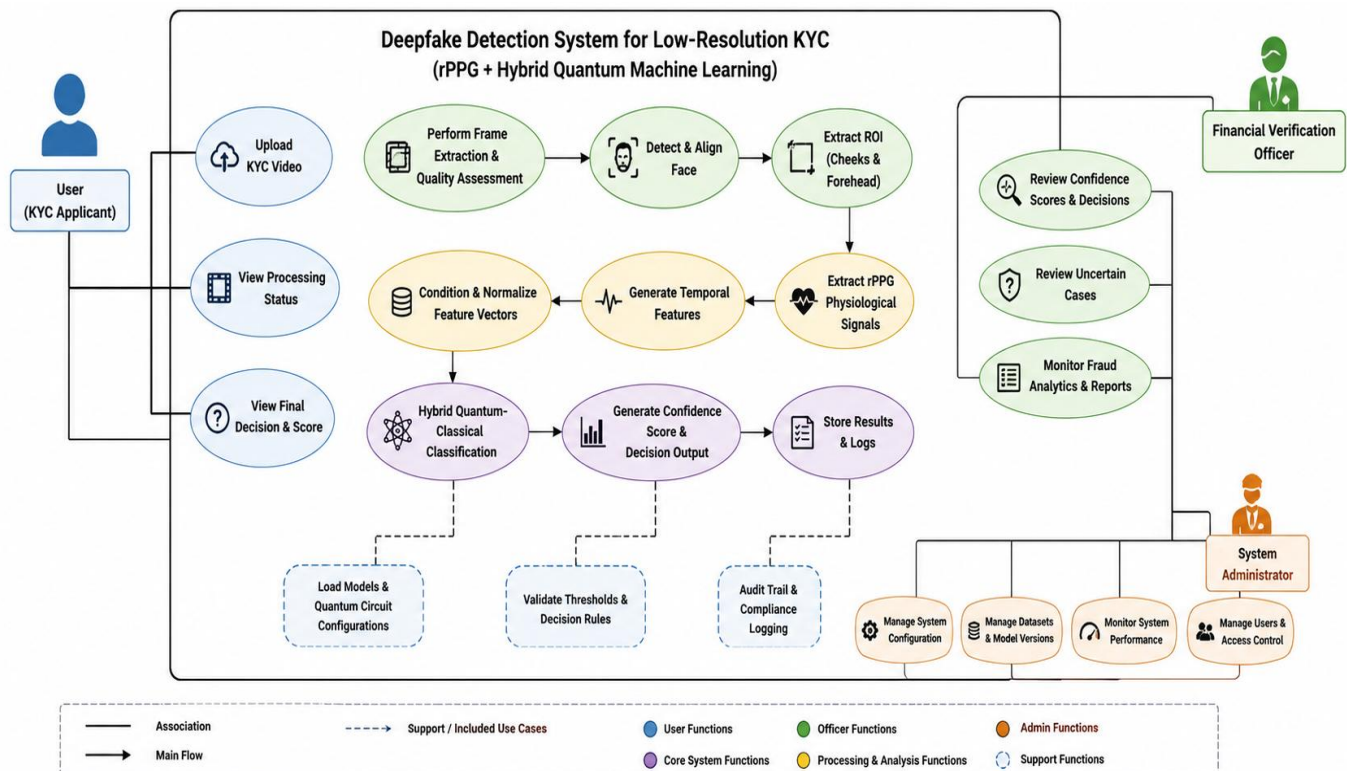


Figure 6.4 – Use Case Diagram

6.2.4 ER Diagram

The Entity Relationship (ER) Diagram represents the data-level organization of the proposed deepfake detection framework. The system includes entities such as User, Video, Frame, FeatureVector, PredictionResult, and ReviewLog.

The User entity stores user-related verification information associated with uploaded KYC videos. The Video entity manages uploaded video metadata and processing status. The Frame entity stores extracted frame sequences associated with a particular video instance. The FeatureVector entity contains physiological and temporal features generated from rPPG analysis. PredictionResult stores classification outputs and confidence scores generated by the hybrid quantum classifier. The ReviewLog entity maintains records related to uncertain predictions and manual verification activities. The relationships between these entities enable structured management of preprocessing results, feature generation outputs, classification decisions, and verification records within the complete authentication workflow.

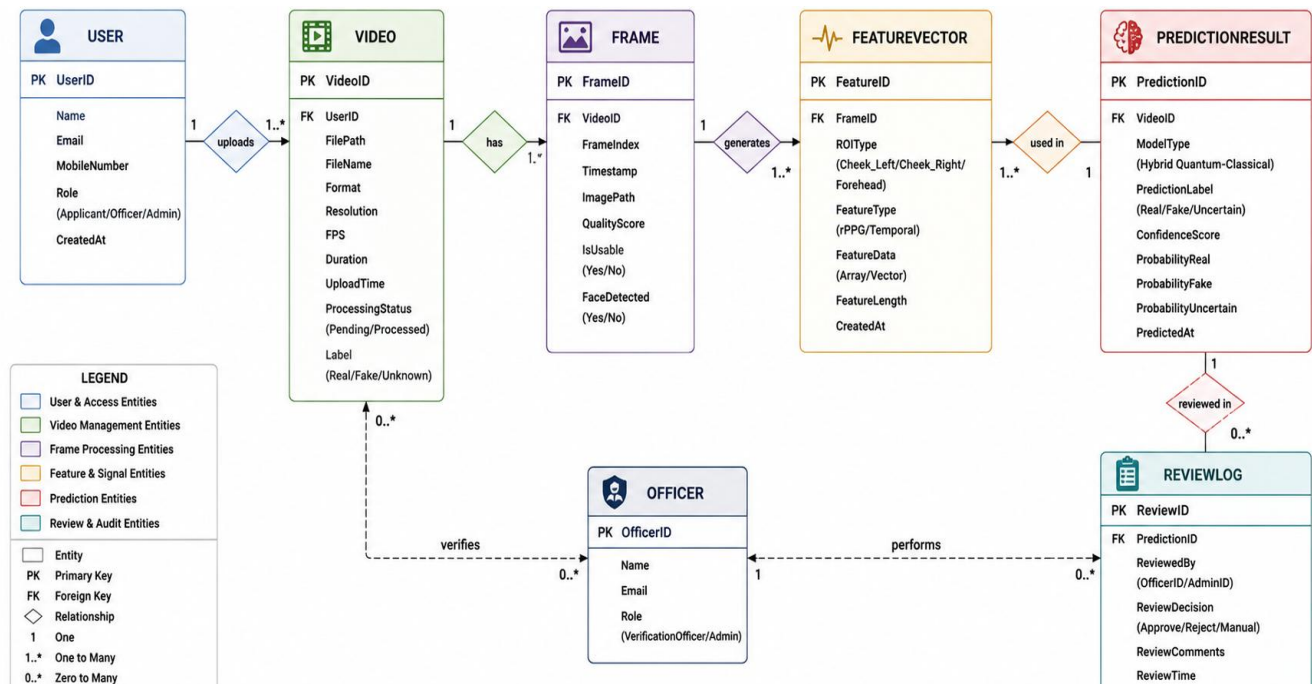


Figure 6.5 – ER Diagram

6.2.5 Scenarios

The system scenarios describe practical operational situations in which the proposed framework performs financial KYC deepfake detection tasks. The scenarios include video upload and validation, frame quality filtering, physiological signal extraction, hybrid classification processing, confidence-aware verification, and manual review support for uncertain outputs.

These scenarios demonstrate how the proposed architecture interacts with users and verification officers while maintaining secure and reliable authentication support under low-resolution KYC conditions.

Scenario ID	Scenario Name	Actor(s) Involved	Scenario Flow (High Level)	System Response / Action	Expected Output	Outcome
S1	Video Upload and Validation	User		- Validate file format, size, duration, frame rate - Check readability and face presence - Accept or reject video		
S2	Frame Quality Filtering	System		- Extract frames at controlled rate - Assess blur, brightness, noise and face visibility - Remove unusable frames		
S3	Physiological Signal Extraction (rPPG)	System		- Detect and align face - Extract ROI for rPPG - Generate temporal physiological features		
S4	Hybrid Quantum-Classical Classification	System		- Normalize and condition features - Apply hybrid quantum-classical model - Generate confidence scores		
S5	Confidence-Aware Verification	Verification Officer		- Compare confidence with defined thresholds - Categorize output accordingly - Auto-approve or flag for review		
S6	Manual Review for Uncertain Cases	Verification Officer		- Review uncertain cases manually - Analyze signals, frames, and scores - Final decision & log action		

Legend:
 User
 Verification Officer
 System Process
 Successful Outcome
 Conditional Outcome
 Manual Outcome

Table 6.1 – System Scenarios

CHAPTER 7

CONCLUSION

The project presents a robust deepfake detection framework designed specifically for low-resolution financial e-KYC and KYC verification environments using rPPG-based physiological signal analysis combined with hybrid quantum machine learning techniques. Unlike conventional deepfake detection approaches that rely mainly on visible facial artifacts or image inconsistencies, the proposed system focuses on biological realism, temporal consistency, and physiological authenticity, which are significantly more difficult to manipulate convincingly in forged videos. This improves the reliability of verification even when the input video quality is degraded because of compression, bandwidth limitations, camera noise, or low-cost mobile devices commonly used in real-world financial authentication systems.

The developed architecture integrates multiple processing stages including video preprocessing, facial stabilization, region of interest extraction, physiological signal generation, compact feature conditioning, and hybrid quantum-enhanced classification into a unified and structured verification pipeline. The incorporation of temporal physiological analysis allows the framework to identify subtle inconsistencies in blood-flow-related facial variations that are generally absent or improperly synthesized in manipulated content. In addition, the use of hybrid quantum-inspired learning mechanisms contributes toward improved feature representation and enhanced decision-making capability while maintaining computational efficiency for lightweight deployment scenarios.

The proposed system also supports confidence-aware authentication outputs that are suitable for practical banking and financial workflows where reliability, explainability, and fraud prevention are critical requirements. By combining physiological signal analysis with advanced machine learning strategies, the framework demonstrates strong potential for improving trustworthiness and security in digital identity verification systems. Furthermore, the modular architecture of the framework enables future scalability and integration with existing e-KYC infrastructures without major architectural modifications.

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